Integration (Riemann)

math-foundations concept 15

1 Motivation

The Riemann integral assigns to a bounded function $f : [a, b] \to \mathbb{R}$ a single number $\int_a^b f$ that represents the signed area between the graph of f and the x-axis. The construction is purely metric: chop $[a, b]$ into finitely many subintervals, approximate f on each subinterval by its supremum (overestimate) and infimum (underestimate), and demand that the two estimates converge as the chopping becomes finer.

2 Partitions and Darboux Sums

Definition 1 (Partition). *A* partition *of* $[a, b]$ *is a finite ordered set* $P = \{x_0, x_1, \ldots, x_n\}$ *with*

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Set $\Delta x_i = x_i - x_{i-1}$ *and define the* mesh $\|P\| = \max_i \Delta x_i$. *A partition* P' refines P *if* $P \subseteq P'$.

Definition 2 (Upper and lower sums). *Let* $f : [a, b] \to \mathbb{R}$ *be bounded. For a partition* P *as above, set*

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x), \qquad m_i = \inf_{x \in [x_{i-1}, x_i]} f(x).$$

The upper Darboux sum *and* lower Darboux sum *are*

$$U(f, P) = \sum_{i=1}^{n} M_i \, \Delta x_i, \qquad L(f, P) = \sum_{i=1}^{n} m_i \, \Delta x_i.$$

Lemma 1. *If* P' *refines* P*, then* $L(f, P) \le L(f, P') \le U(f, P') \le U(f, P)$*. Consequently* $L(f, P_1) \le U(f, P_2)$ *for any two partitions* P_1, P_2.

Definition 3 (Darboux integrability). *Define the* upper *and* lower integrals

$$\overline{\int_a^b} f = \inf_P U(f, P), \qquad \underline{\int_a^b} f = \sup_P L(f, P).$$

We say f *is* Riemann (Darboux) integrable *on* $[a, b]$ *if the two coincide; the common value is denoted* $\int_a^b f(x)\, dx$.

Proposition 2 (Cauchy criterion). *A bounded* f *is integrable on* $[a, b]$ *iff for every* $\varepsilon > 0$ *there exists a partition* P *with* $U(f, P) - L(f, P) < \varepsilon$.

Proof. ($\Rightarrow$) Suppose $\overline{\int} f = \underline{\int} f =: I$. By the definitions of inf and sup, pick P_1, P_2 with $U(f, P_1) < I + \varepsilon/2$ and $L(f, P_2) > I - \varepsilon/2$. The common refinement $P = P_1 \cup P_2$ satisfies $U(f, P) \le U(f, P_1)$ and $L(f, P) \ge L(f, P_2)$, so $U(f, P) - L(f, P) < \varepsilon$.

($\Leftarrow$) For every ε we have $0 \le \overline{\int} f - \underline{\int} f \le U(f, P) - L(f, P) < \varepsilon$. Hence the difference is 0. $\square$

3 Continuity Implies Integrability

Theorem 3. *Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann integrable.*

Proof. Because $[a, b]$ is compact, f is bounded and uniformly continuous (Heine–Cantor). Fix $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ such that

$$x, y \in [a, b],\ |x - y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{b - a}.$$

Take any partition $P = \{x_0, \ldots, x_n\}$ with mesh $\|P\| < \delta$.

On each subinterval $[x_{i-1}, x_i]$, the function f attains its sup M_i and inf m_i (continuous on a compact set), say at points $\xi_i, \eta_i \in [x_{i-1}, x_i]$. Since $|\xi_i - \eta_i| \leq \Delta x_i < \delta$,

$$M_i - m_i = f(\xi_i) - f(\eta_i) < \frac{\varepsilon}{b - a}.$$

Summing over i,

$$U(f, P) - L(f, P) = \sum_{i=1}^{n} (M_i - m_i)\, \Delta x_i < \frac{\varepsilon}{b - a} \sum_{i=1}^{n} \Delta x_i = \frac{\varepsilon}{b - a} \cdot (b - a) = \varepsilon.$$

By Proposition 2, f is integrable. □

4 The Fundamental Theorem of Calculus

Theorem 4 (FTC, Part 1). *Let $f : [a, b] \to \mathbb{R}$ be continuous, and define $F : [a, b] \to \mathbb{R}$ by $F(x) = \int_a^x f(t)\, dt$. Then F is differentiable on (a, b) and $F'(x) = f(x)$.*

Sketch. For $h \neq 0$, $\frac{F(x+h)-F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t)\, dt$. Bound the integrand by $f(x) \pm \varepsilon$ using continuity of f at x, and let $h \to 0$. □

Theorem 5 (FTC, Part 2). *If f is continuous on $[a, b]$ and G is any antiderivative of f, then $\int_a^b f(t)\, dt = G(b) - G(a)$.*

Sketch. Both F from Part 1 and G are antiderivatives of f, so they differ by a constant: $G(x) = F(x) + C$. Then $G(b) - G(a) = F(b) - F(a) = \int_a^b f - 0$. □

5 Worked Example

For $f(x) = x^2$ on $[0, 1]$ with equal partition $x_i = i/n$,

$$U(f, P_n) = \sum_{i=1}^{n} \frac{i^2}{n^2} \cdot \frac{1}{n} = \frac{(n+1)(2n+1)}{6n^2} \xrightarrow[n \to \infty]{} \frac{1}{3}.$$

By FTC2 the antiderivative $x^3/3$ also yields $1/3 - 0 = 1/3$.